

Lecture 1

Introduction to SQA

Outline

- What is Software?
- What is SQA?
- SQA Standards

References

- Stair & Reynolds, Principles of Information Systems, chapter 4
- Daniel Galin, Software Quality Assurance: From theory to implementation, chapter 2.1–2.5

What is Software?

A program vs A software

- **A program:** A program can run in computer, phone or machine.
- **A software:** Computer programs and all associated documentation and configuration data. Software products may be developed for a particular customer or may be developed for a general market.

What is Software?

***“Software is:** Computer programs, procedures, and possibly associated documentation and data pertaining to the operation of a computer system.”*

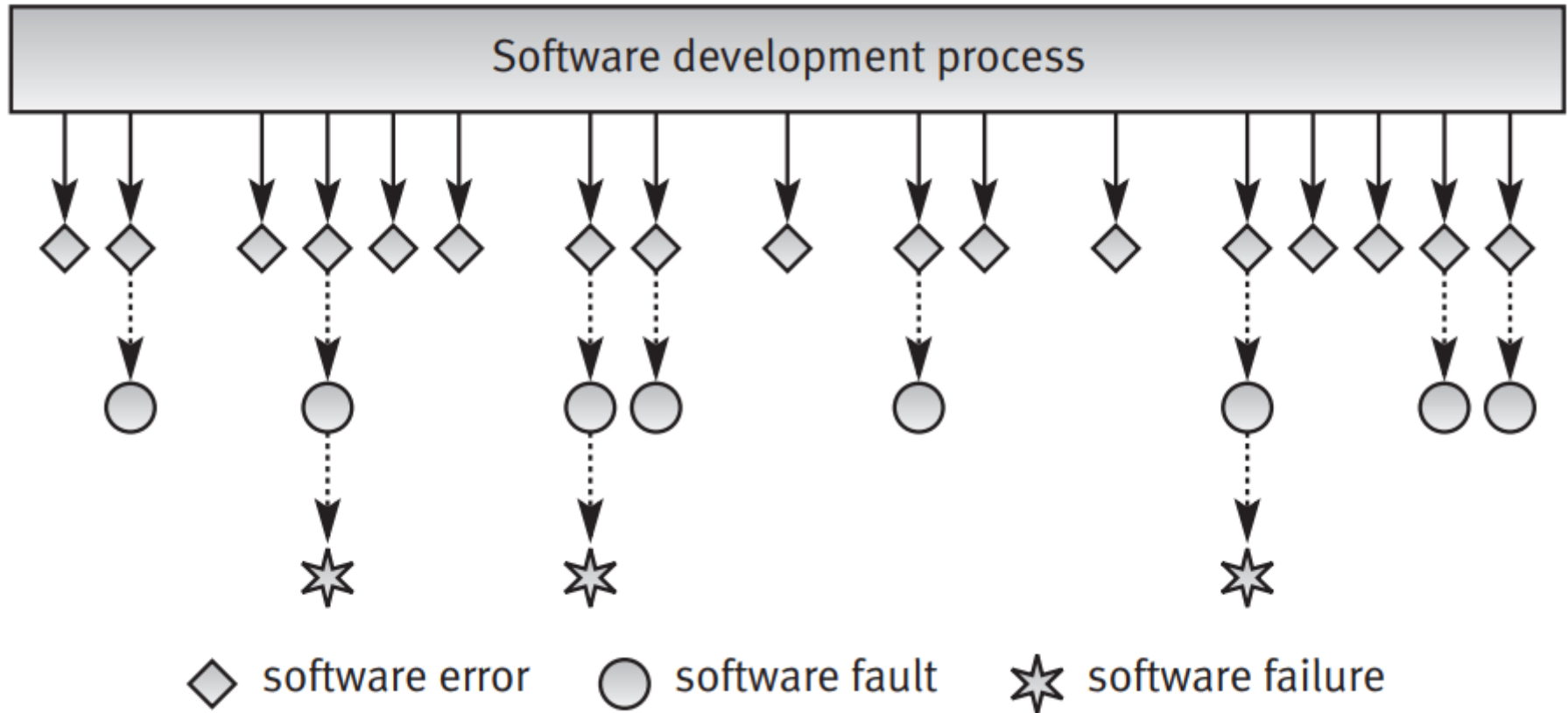
Core Components:

- Computer programs (the “code”)
- Procedures
- Documentation
- Data necessary for operating the software system

Software errors, faults and failures

- **Software Error – made by programmer**
 - Syntax (grammatical) error
 - Logic error in carrying out client's requirements (multiply vice add two operands)
- **Software Fault**
 - All software errors may not cause software faults
 - That part of the software may not be executed
 - (An error is present but not encountered....)
- **Software Failures**
 - A software fault transforms into a software failure only when it is activated

Software errors, faults and failures



Causes of software errors

1. Faulty requirements definition

- **Erroneous/Incomplete:** Requirements are defined incorrectly or vital parts are missing (e.g., forgetting a tax discount for students).
- **Absence of requirements:** Vital system functions are simply not documented.
- **Unnecessary requirements:** Including functions that are not needed, adding unhelpful complexity.

Causes of software errors

2. Client–Developer Communication Failures

- **Misunderstanding:** Misinterpreting written or oral instructions/changes from the client.
- **Poor response handling:** Misunderstanding the client's feedback on design problems.
- **Lack of attention:** Failing to track or record client messages regarding requirement updates.

Causes of software errors

3. Deliberate Deviations from Requirements

- **Improper Reuse:** Using old software modules without adapting them to new requirements.
- **Pressure Shortcuts:** Omitting required functions due to time or budget constraints.
- **Unapproved "Improvements":** Developers adding "minor" features without client approval, often causing hidden errors.

Causes of software errors

4. Logical Design Errors

- **Erroneous Algorithms:** Representing requirements through flawed mathematical or logic flows.
- **Process Sequencing:** Incorrect order of operations (e.g., skipping a manual approval step).
- **Boundary Conditions:** Errors in defining limits (e.g., using "3 or more" instead of "more than 3").
- **Omissions:** Failing to define reactions for specific system states or illegal user operations (causing crashes).

Causes of software errors

5. Coding Errors

- **Programming Flaws:** Misunderstanding design documents or making linguistic/syntax errors.
- **Tool/Data Errors:** Improper use of CASE tools or incorrect data selection.
- **Common Types:** Syntax errors, Logic errors (wrong results), and Run-time errors (crashes).

Causes of software errors

6. Non-compliance with Instructions

- **Coordination Difficulties:** Team members struggle to integrate code from "non-complying" peers.
- **Maintenance Hurdles:** Difficulty for new members or maintenance teams to understand undocumented/non-standard code.
- **Testing Inefficiency:** Test teams find it harder to detect and correct errors in non-standard modules.

Causes of software errors

7. Shortcomings of the Testing Process

- **Incomplete test plans:** Parts of application not tested or tested thoroughly!
- **Failure:**
 - to document, report detected errors and faults
 - to promptly correct detected software faults as a result of inappropriate indications of the reasons for the fault.
- **Incomplete correction of detected errors** due to time pressures.

Causes of software errors

8. Procedure and Documentation Errors

- **Procedure Errors:** Providing incorrect directions to users for complex, multi-step processes.
- **Documentation Omissions:** Omission of software functions in user manuals.
- **Instruction Errors:** Giving incorrect explanations that lead to "dead ends" for the user.
- **False Listings:** Including functions that were planned but later cancelled or dropped.

What is SQA?

Software quality – definition

Software quality is:

1. The degree to which a system, component, or process meets **specified requirements** (*Crosby, 1979*)

Deficiency

???

Software quality – definition

Software quality is:

2. The degree to which a system, component, or process meets **customer needs or expectations**
(Juran)

Deficiency

???

Software quality – definition

***Software quality is:** Conformance to explicit requirements, documented standards, and implicit characteristics (Pressman, 2000)*

Three requirements:

- **Explicit Requirements:** Functional and performance goals
- **Documented Standards:** Specific quality standards mentioned in the contract
- **Implicit Characteristics: Good Software Engineering Practices (GSEP)** expected of all professional software (e.g., maintainability, readability)

Software Quality Assurance

Software quality assurance is:

1. A **planned and systematic pattern** of all **actions** necessary to provide adequate **confidence** that an item or product conforms to established technical requirements.
2. A **set of activities** designed to evaluate the **process** by which the products are developed or manufactured. Contrast with quality control.

SQA – Expanded definition

Software quality assurance is: A systematic, planned set of actions necessary to provide adequate confidence that the software development process or the maintenance process of a software system product conforms to established functional technical requirements as well as with the managerial requirements of keeping the schedule and operating within the budgetary confines

Covers both the Development and Maintenance processes throughout the lifecycle

Software quality assurance vs. software quality control

Quality Control is defined as a set of activities designed to evaluate the quality of a developed or manufactured product

- We have QC inspections during **development** and before deployment
- QC activities are only a part of the total range of QA activities.

Software quality assurance vs. software quality control

Quality Assurance's objective is to minimize the **cost of guaranteeing quality** by a **variety of activities performed throughout the development/manufacturing processes/stages**

- Activities prevent causes of errors; detect and correct them early in the development process
- QA substantially reduces the rate of products that do not qualify for shipment and/at the same time, reduce the costs of guaranteeing quality in most cases.

The objectives of SQA activities

Software development (process-oriented):

1. Assuring an acceptable level of confidence that the software will conform to functional technical requirements
2. Assuring an acceptable level of confidence that the software will conform to managerial scheduling and budgetary requirements
3. Initiation and management of activities for the improvement and greater efficiency of software development and SQA activities

The objectives of SQA activities

Software maintenance (product-oriented):

1. Assuring with an acceptable level of confidence that the software maintenance activities will conform to the functional technical requirements.
2. Assuring with an acceptable level of confidence that the software maintenance activities will conform to managerial scheduling and budgetary requirements.
3. Initiating and managing activities to improve and increase the efficiency of software maintenance and SQA activities.

SQA Standards

Standards are external tools used to integrate international expertise into an organization's quality system

- **Utilization:** Leveraging global professional knowledge and best practices
- **Coordination:** Improving alignment with the quality systems of partners and other organizations
- **Objective Evaluation:** Providing an independent, professional framework to measure quality achievements

Classification of SQA Standards

Quality Management Standards (Focus: "The WHAT")

- **Purpose:** Guides the management of development, maintenance, and infrastructure.
- **Approach:** Focuses on **what is required**, leaving the "how to achieve it" to the organization.
- **Certification:** Compliance allows organizations to seek official SQA certification.
- **Examples:** ISO 9001, ISO 9000-3, SEI CMM.

Classification of SQA Standards

Project Process Standards (Focus: "The HOW")

- **Purpose:** Provides the development team with specific methodological guidelines.
- **Approach:** Focuses on **how to perform** specific tasks and procedures during the project.
- **Examples:** IEEE 1012 (V&V), ISO/IEC 12207 (Software Life Cycle Processes).

Summary

- What is Software?
- What is Error, Fault, Failure?
- What is SQA?
- SQA Standards